HAMED HOJATIAN

Ph.D. Candidate

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EDUCATION

Ph.D. in Applied Deep Learning in wireless communication Systems

Polytechnique Montréal

Fall 2019 - current — Montreal, Canada

Supervisor: **Prof. François Leduc-Primeau**
Co-Supervisor: **Prof. Jean-François Frigon**
CGPA: **4.0/4.0**

M.Sc. in Telecommunication Engineering

Isfahan University of Technology (IUT)

2014 - 2017 — Isfahan, Iran

Supervisor: **Prof. Mohammad Javad Omidi**
CGPA: **3.5/4.0**

B.Sc. in Electrical Engineering

Babol (Noshirvani) University of technology (NIT)

2010 - 2014 — Mazandaran, Iran

Supervisor: **Prof. Hosein Miar-Naeimi**
CGPA: **3.1/4.0**

DISSERTATION WORK

Ph.D. Thesis

Hybrid Beamforming in Massive MIMO Systems with Deep Learning

- RSSI-Based hybrid beamforming design with deep learning;
- Improve the spectral and energy efficiency of hybrid beamforming design;
- Unsupervised learning for hybrid beamforming design;
- Coordinated hybrid beamforming and cell-free massive MIMO with deep neural network; and
- ...

M.Sc. Thesis

Time and Frequency synchronization in Massive MIMO systems

- Carrier Frequency Offset (CFO) and channel estimation Using ZC sequence;
- Detection in receiver using iterative method; and
- Improve the method using Filtered- OFDM.

B.Sc. Thesis

OFDM Transmitter and receiver

- Developed in MATLAB simulation
- OFDMA multi-user simulation in AWGN channel.

SEEKING FOR

"To work in a progressive and dynamic research organization where one could solve challenging scientific problems"

PROG. SKILLS

- Python
- Pytorch
- JAVA SCRIPT
- MATLAB
- LaTeX
- MS Office
- HTML5
- CSS
- Visual basic
- C / C++
- Bash

LANGUAGES

- English
- French
- Persian

TELECOMM. TOOLS

- Ericsson OSS
- MapInfo
- Business Object (BO)
- Atoll
- TEMS
- Schema
- Actix

STRENGTHS

Hard-working | Decision Making | Ability to Work Under Pressure | Motivator & Leader | Conflict Resolution | Time Management | Target oriented | Flexibility and Adaptability

RESEARCH INTERESTS

- Applied ML in telecommunication;
- Deep Learning;
- Reinforcement learning;
- Signal processing in MASSIVE MIMO communication;
- Millimeter wave in Massive MIMO and Antenna array;
- Coordinated, cooperative beamforming and distributed antenna; and
- Cell free communication.

PUBLICATION

- H. Hojatian, J. Nadal, J.-F. Frigon, and F. Leduc-Primeau. **"Unsupervised Deep Learning for Massive MIMO Hybrid Beamforming"**, IEEE Transactions on Wireless Communications, arXiv:2007.00038. (2nd review round)
- H. Hojatian, V. N. Ha, J. Nadal, J.-F. Frigon, and F. Leduc-Primeau. **"RSSI-Based Hybrid Beamforming Design with Deep Learning"**, 2020 IEEE International Conference on Communications (ICC): Wireless Communications Symposium (IEEE ICC'20 - WC Symposium), Dublin, Ireland, 2020.
- H. Hojatian, M. J. Omidi, H. Saeedi-Sourk, A. Farhang, **"Joint CFO and channel estimation in OFDM-based massive MIMO systems"**, 2016 8th International Symposium on Telecommunications (IST), Tehran, 2016, pp. 343-348, doi: 10.1109/ISTEL.2016.7881837.

WORK EXPERIENCE

Senior Telecommunication Engineer

Mobile Telecommunication Company of Iran (MCI)

Nov 2015 - Nov 2018 | Tehran, Iran

- Ericsson OSS in Tehran Network;
- Network Planning (2G, 3G and LTE);
- Network Optimization (BSC owner and RNC owner);
- Project manager of RNC rehoming project in Tehran; and
- Parameter tuning.

Vice-President

Eleventh (11th) national robotics competition at BABOL University of Technology

Oct 2015 | Mazandaran, Iran

CERTIFICATES

- "**4TH IVADO/MILA SCHOOL IN DEEP LEARNING**" *IVADO/MILA*, 2019, Montreal, Canada.
- "**DEEP LEARNING WITH PYTORCH**" *Udemy*, 2018, Online.
- "**LTE OPTIMIZATION OF RADIO NETWROK (3GPP)**" *TECHCOM Consulting GmbH*, 2018, Tehran, Iran.
- "**LTE COUNTER AND KPI (ERICSSON)**" *TECHCOM Consulting GmbH*, 2017, Tehran, Iran.
- "**LTE RADIO NETWORK PLANNING**" *TECHCOM Consulting GmbH*, 2017, Tehran, Iran.

TALKS & TEACHING

Author

IEEE ICC2020

Jun 2020 | Virtual-Dublin, Ireland

- Present my accepted paper in ICC2020;

Teaching assistant

BNUT

2013 | Mazandaran, Iran

- Teaching assistant for Electronic I
- Teaching assistant for Electronic circuit I
- Teaching assistant for Digital Signal Processing

REFEREES

Dr. François Leduc-Primeau
@ francois.leduc-primeau@polymtl.ca
Assistant Professor, Department of EE, Polytechnique Montréal.

Dr. Jean-François Frigon
@ jf.frigon@polymtl.ca
Full Professor, Department of EE, Polytechnique Montréal.

Dr. Mohammad Javad Omidi
@ omidi@cc.iut.ac.ir
Assistant Professor, Department of EE, Isfahan University of Technology.@

HOBBIES

Music
Listen quality music of every genre

Searching
Always eager to find the updated topics in my fields

Swimming
Fresh water is my charger

COURSES

Polytechnique Montréal (PolyMTL)

- Learning Representation (A+) - instructor: Prof. Aaron Courville
- Wireless Communication (A) - instructor: Prof. Jean-François Frigon
- Estimation and Detection Theory (A) - instructor: Prof. Jean-François Frigon

Isfahan University of Technology (IUT)

- Advance Digital Communication (ADC)
- Information Theory
- Software Define Radio (SDR)
- Digital Signal Processing (DSP)